mathematically inclined. But, even if you're a physicist, what's the virtue in shutting up and calculating? Mermin himself provided the answer in his 1989 article. "It is a fact about the quantum theory of paramount importance which ought to be emphasized in every popular and semi-popular exposition, that it permits us to calculate measurable quantities with unprecedented precision." Quantum physics works. The calculations enabled by the theory are astonishing in their range of applicability and the accuracy of their results. Quantum physics tells us how long it will take to heat up your frying pan to cook your eggs and how large a dying white dwarf star can be without collapsing. It reveals the exact shape of the double helix at the core of life, it tells us the age of the immortal cattle on the rock walls at Lascaux, it speaks of atoms split beneath the stone heart of Africa eons before Oppenheimer and the blinding light of Trinity. It predicts with uncanny accuracy the precise darkness of the blackest night. It shows us the history of the universe in a handful of dust. If shutting up is the price of doing these calculations, then pass the ball gag and break out the graph paper.

But why is that the price? Why does Copenhagen require that you shut up in order to calculate? For that matter, how does the Copenhagen interpretation allow you to calculate at all? The measurement problem is so centrally tied to the core of quantum physics that, without some answer to the problem, it's impossible to use the theory. Some interpretation must guide you in the use of the mathematics—and Copenhagen, as we've seen many times over, offers no such solution and is not a true interpretation. So how can shutting up allow you to calculate anything?

The form of the Copenhagen interpretation usually found in physics textbooks says (explicitly or otherwise) that measurement is a fundamentally different process from any other found in nature and that "measurement" is defined as "any time a large object encounters a small one." Large objects are simply assumed to obey classical physics, even as quantum physics is presented to the student as a more fundamental theory underpinning classical physics. In short, the student is implicitly asked to accept as part of the basic structure of quantum physics that there are two worlds, the classical and the quantum, just as Bohr taught. Yet, at the same time, they're being told that quantum physics is the fundamental theory from which classical physics emerges. So quantum physics students are